CO2 PRACTICAL EXERCISE

1. Zero-shot, One-shot, and Few-shot Prompting for a Smart Fitness Watch Product Description

CODE:

```
prompts = {
    "Zero-shot": """
Write a product description for a Smart Fitness Watch.
Highlight its fitness tracking, heart-rate monitoring,
sleep tracking, and battery life.
""",

    "One-shot": """
Example:
Product: Wireless Earbuds
Description: Enjoy clear sound, comfortable fitting,
and reliable wireless connectivity for everyday use.

Product: Smart Fitness Watch
Write a similar product description highlighting
fitness tracking, heart-rate monitoring, and sleep tracking.
""",

    "Few-shot": """
Example 1:
Product: Smart Water Bottle
Description: Track your hydration and build healthier
drinking habits with convenient reminders.

Example 2:
Product: Fitness Band
Description: Monitor your daily steps, activity levels,
and sleep to support a healthier lifestyle.

Product: Smart Fitness Watch
```

Write a similar description highlighting fitness tracking,

heart-rate monitoring, sleep tracking, and battery life.

"""

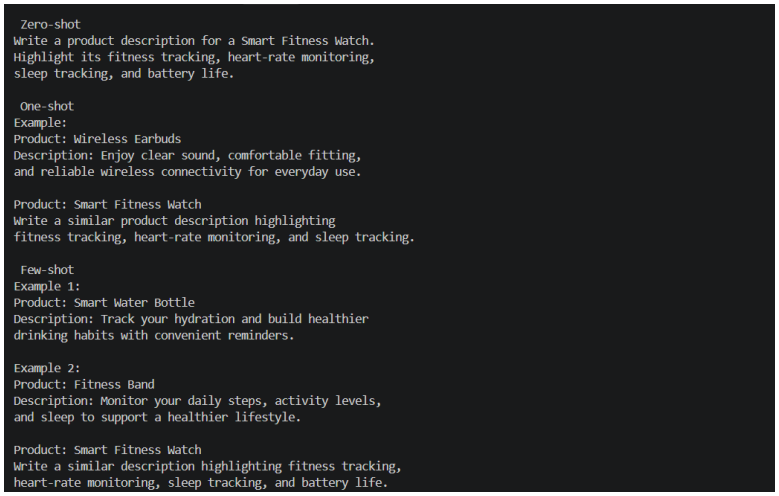
}

for method, prompt in prompts.items():

print("\n", method)

print(prompt.strip())

OUTPUT:



2. Generate a 200-word Healthcare AI Blog Using Zero-shot, One-shot, and Few-shot Prompting

CODE:

prompts = {

"Zero-shot": """

Write a blog of approximately 200 words titled

'Applications of Artificial Intelligence in Healthcare'.

Include diagnosis, medical imaging, drug discovery,

patient monitoring, benefits, and challenges.

""",

"One-shot": """

Example:

Topic: AI in Education

```
Structure: Introduction, applications, benefits,
challenges, conclusion.

Now write approximately 200 words on
'Applications of Artificial Intelligence in Healthcare'
using the same structure.
""",

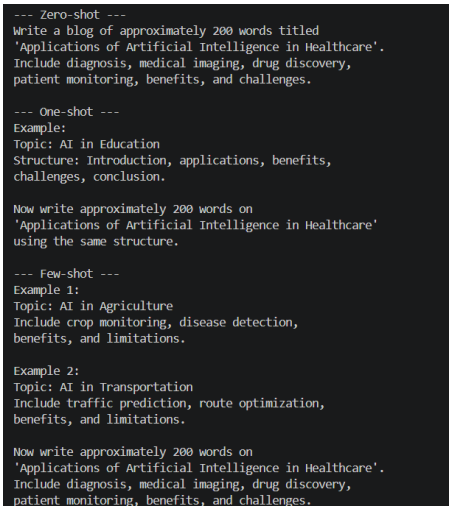
    "Few-shot": """
Example 1:
Topic: AI in Agriculture
Include crop monitoring, disease detection,
benefits, and limitations.

Example 2:
Topic: AI in Transportation
Include traffic prediction, route optimization,
benefits, and limitations.

Now write approximately 200 words on
'Applications of Artificial Intelligence in Healthcare'.
Include diagnosis, medical imaging, drug discovery,
patient monitoring, benefits, and challenges.
"""
}

for method, prompt in prompts.items():
    print("\n---", method, "---")
    print(prompt.strip())
```

OUTPUT:



3. Summarize an Article in 50 Words Using Different Prompting Techniques

CODE:

```
article = """
Artificial Intelligence is increasingly used in healthcare.
It supports medical image analysis, disease prediction,
patient monitoring, and drug discovery. AI tools can help
doctors analyze large amounts of data quickly. However,
healthcare AI must protect patient privacy, reduce bias,
and be validated by medical professionals before use.
"""

prompts = {
    "Zero-shot": f"""
Summarize the following article in exactly 50 words.
Preserve its main ideas and do not add unsupported facts.

Article: {article}
""",

    "One-shot": f"""
Example:
Article: Solar energy converts sunlight into electricity.
It reduces dependence on fossil fuels but depends on
```

```
weather and storage technology.

Summary: Solar energy produces electricity from sunlight,
reducing fossil-fuel dependence. Its effectiveness depends
on weather conditions and energy-storage technology.

Now summarize this article in exactly 50 words:
{article}
""",

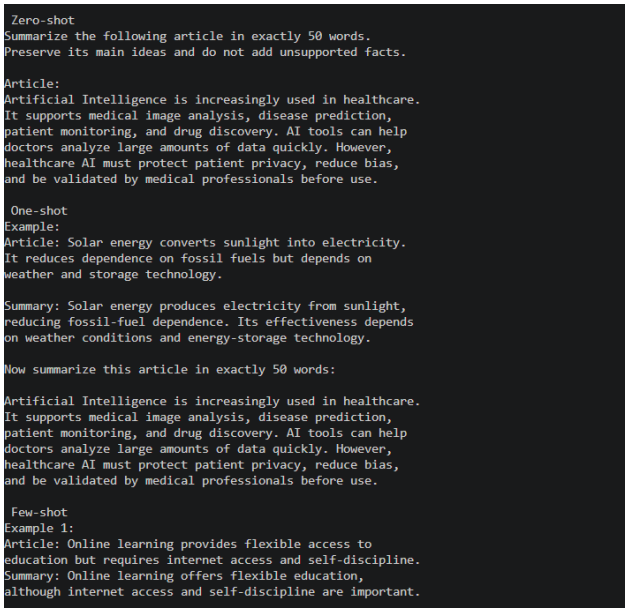
    "Few-shot": f"""
Example 1:
Article: Online learning provides flexible access to
education but requires internet access and self-discipline.
Summary: Online learning offers flexible education,
although internet access and self-discipline are important.

Example 2:
Article: Recycling reduces waste and conserves resources,
but requires effective collection and sorting.
Summary: Recycling conserves resources and reduces waste,
provided collection and sorting systems work effectively.

Summarize this article in exactly 50 words:
{article}
"""
}

for method, prompt in prompts.items():
    print("\n", method)
    print(prompt.strip())
```

OUTPUT:



4. Generate a Professional Leave Email Using Different Prompting Techniques

CODE:

prompts = {

"Zero-shot": """

Write a professional email to my manager requesting

two days of sick leave due to illness.

Include a subject, greeting, reason, leave dates,

polite request, and closing.

""",

"One-shot": """

Example:

Subject: Request for Personal Leave

Dear Manager,

I would like to request personal leave for one day

on 12 November. Please let me know if approval

is possible.

Thank you,

Employee

Now write a professional sick-leave email requesting

two days off due to illness. Use placeholders for dates.

""",

"Few-shot": """

Example 1:

Subject: Request for Leave

Dear Sir/Madam,

I am requesting leave on [date] due to a personal

commitment. Thank you for considering my request.

Regards, [Name]

Example 2:

Subject: Medical Appointment Leave

Dear Manager,

I request permission to leave early on [date]

for a medical appointment. I will complete my

pending tasks accordingly.

Regards, [Name]

Now write a sick-leave email requesting two days off.

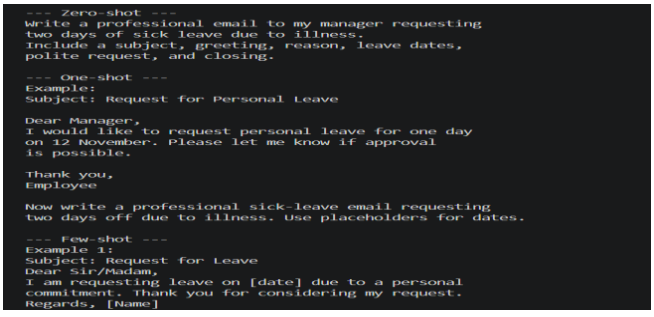
Include the subject, reason, dates, and professional closing.

"""

}

```
for method, prompt in prompts.items():
    print("\n---", method, "---")
    print(prompt.strip())
```

OUTPUT:



5. Generate a Promotional Social Media Post for an AI Workshop

CODE:

```
prompts = {
    "Zero-shot": """
Create an attractive social media post promoting an
AI Workshop for college students. Include benefits,
date [date], venue [venue], registration details,
hashtags, and a call to action.
""",

    "One-shot": """
Example:
Join our Python Workshop!
Learn coding basics through practical activities.
Date: [date] | Venue: [venue]
Register now and start coding!
#PythonWorkshop #LearnToCode

Now create a similar promotional post for an AI Workshop.
""",

    "Few-shot": """
Example 1:
Join our Web Development Workshop!
Build your first website and explore modern web tools.
Register today! #WebDevelopment

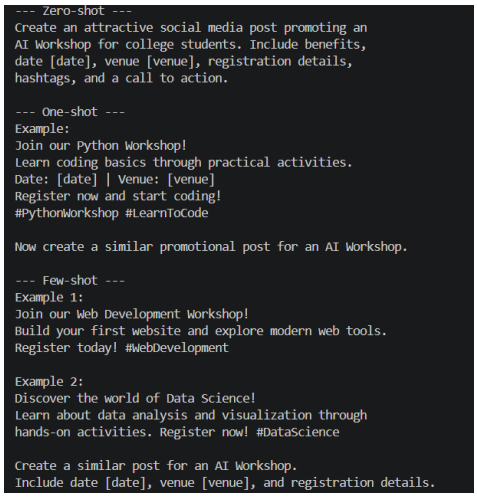
Example 2:
Discover the world of Data Science!
Learn about data analysis and visualization through
hands-on activities. Register now! #DataScience

Create a similar post for an AI Workshop.
Include date [date], venue [venue], and registration details.
```

```
"""
}

for method, prompt in prompts.items():
    print("\n---", method, "---")
    print(prompt.strip())
```

OUTPUT:



6. Text Generation Using the OpenAI LLM API

CODE:

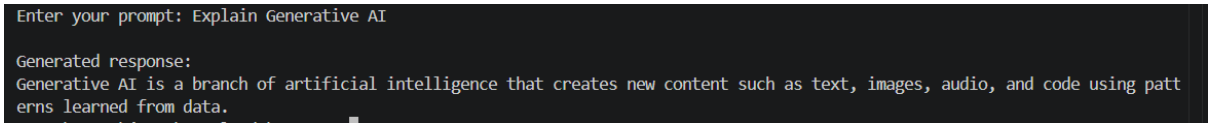
```
prompt = input("Enter your prompt: ")


if "generative ai" in prompt.lower():
    response = (
        "Generative AI is a branch of artificial intelligence "
        "that creates new content such as text, images, audio, "
        "and code using patterns learned from data."
    )
elif "healthcare" in prompt.lower():
    response = (
        "AI in healthcare supports medical image analysis, "
        "disease prediction, patient monitoring, and drug discovery."
    )
else:
```

```
    response = (
        "This topic can be explored by identifying its meaning, "
        "applications, benefits, and limitations."
    )

print("\nGenerated response:")
print(response)
```

OUTPUT:



7. Text Generation Using the Google Gemini LLM API

CODE:

```
responses = {
    "ai": "Artificial Intelligence enables computers to perform tasks that normally require human intelligence.",
    "education": "AI supports personalized learning, automated assessment, and intelligent tutoring.",
    "healthcare": "AI supports medical imaging, patient monitoring, and disease prediction."
}

prompt = input("Enter your prompt: ").lower()

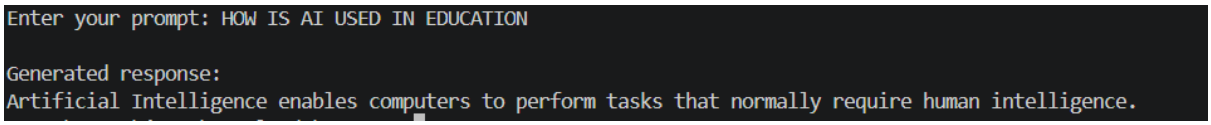
found = False

for topic, answer in responses.items():
    if topic in prompt:
        print("\nGenerated response:")
        print(answer)
        found = True
        break

if not found:
    print("\nGenerated response:")
```

```
    print("Please enter a prompt about AI, education, or healthcare.")
```

OUTPUT:



8. Text Generation Using the Hugging Face Inference API

CODE:

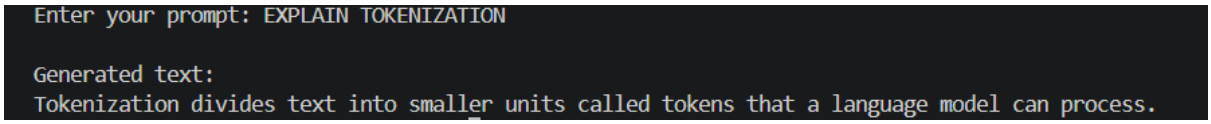
```
def generate_text(prompt):
    prompt = prompt.lower()


    if "machine learning" in prompt:
        return (
            "Machine learning enables computers to learn patterns "
            "from data and make predictions."
        )
    elif "transformer" in prompt:
        return (
            "Transformers process sequences using attention "
            "mechanisms and are widely used in language models."
        )
    elif "tokenization" in prompt:
        return (
            "Tokenization divides text into smaller units called "
            "tokens that a language model can process."
        )
    else:
        return "Enter a topic related to Generative AI."


user_prompt = input("Enter your prompt: ")


print("\nGenerated text:")
print(generate_text(user_prompt))
```

OUTPUT:



9. Python Code Generation Using Structured Prompting

CODE:

```
prompt = """
Role: Python programming assistant
Problem: Determine whether an integer is even or odd.
Input: One integer.
Output: Print Even or Odd.
Constraint: Use the modulo operator.
Return only executable Python code.
"""

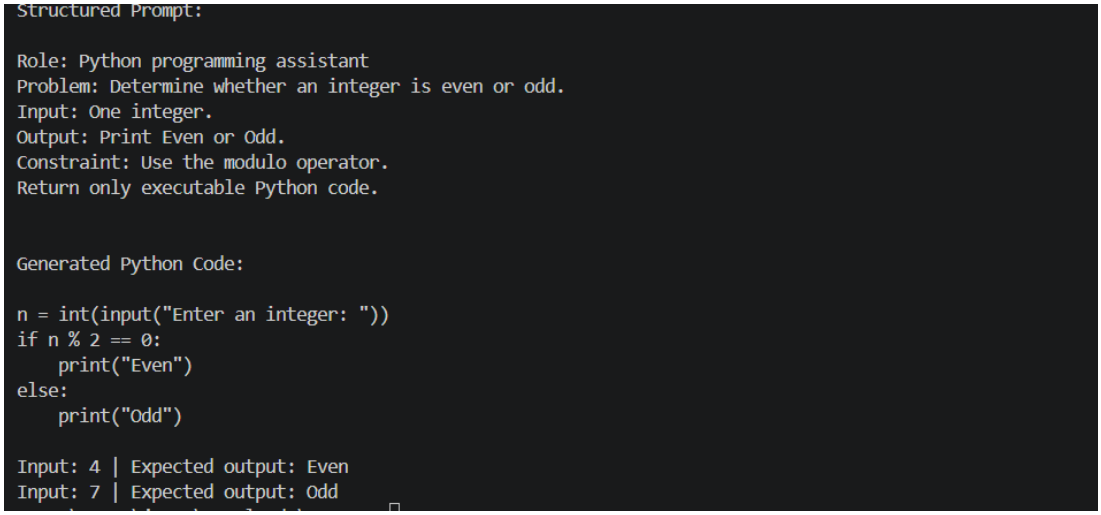
generated_code = """
n = int(input("Enter an integer: "))
if n % 2 == 0:
    print("Even")
else:
    print("Odd")
"""

print("Structured Prompt:")
print(prompt)
print("\nGenerated Python Code:")
print(generated_code)

test_inputs = [4, 7]

for n in test_inputs:
    expected = "Even" if n % 2 == 0 else "Odd"
    print("Input:", n, "| Expected output:", expected)
```

OUTPUT:



10. SQL Query Generation Using Structured Prompting

CODE:

```
schema = """
Table: Students
Columns:
student_id INTEGER
name VARCHAR(50)
department VARCHAR(50)
marks INTEGER
"""

requirement = """
Display the names and marks of students who scored
more than 80 marks, ordered by marks in descending order.
"""

prompt = f"""
Role: SQL query assistant

Database schema:
{schema}
```

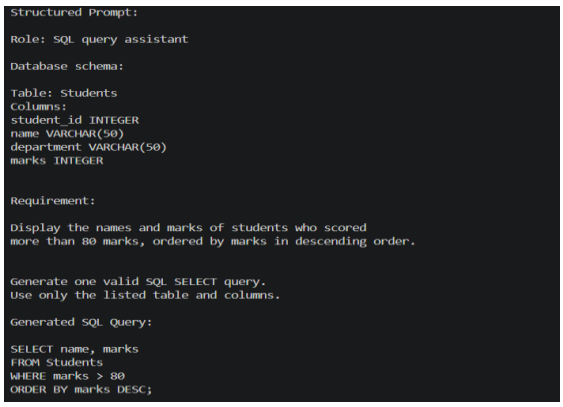
```
Requirement:
{requirement}

Generate one valid SQL SELECT query.
Use only the listed table and columns.
"""

generated_sql = """
SELECT name, marks
FROM Students
WHERE marks > 80
ORDER BY marks DESC;
"""

print("Structured Prompt:")
print(prompt)
print("Generated SQL Query:")
print(generated_sql)
```

OUTPUT:



11. Comparative Analysis of Zero-shot, One-shot, and Few-shot Prompting

CODE:

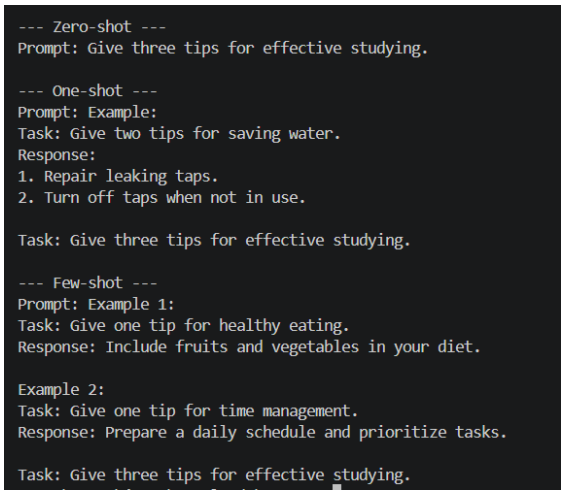
```
task = "Give three tips for effective studying."

prompts = {
    "Zero-shot": task,
```

```
    "One-shot": """
Example:
Task: Give two tips for saving water.
Response:
1. Repair leaking taps.
2. Turn off taps when not in use.
Task: Give three tips for effective studying.
""",
    "Few-shot": """
Example 1:
Task: Give one tip for healthy eating.
Response: Include fruits and vegetables in your diet.
Example 2:
Task: Give one tip for time management.
Response: Prepare a daily schedule and prioritize tasks.

Task: Give three tips for effective studying.
"""
}
for method, prompt in prompts.items():
    print("\n---", method, "---")
    print("Prompt:", prompt.strip())
```

OUTPUT:



12. Prompt Evaluation, Comparison, and Refinement Using LLM Responses

CODE:

```
prompts = {
    "Prompt A": "Explain AI in healthcare.",
    "Prompt B": """
Explain three applications of AI in healthcare.
Give one example for each application.
Use simple language and a numbered list.
"""
}

criteria = [
    "Relevance",
    "Accuracy",
    "Completeness",
    "Clarity",
    "Format adherence"
]

scores = {
    "Prompt A": [4, 4, 3, 4, 3],
    "Prompt B": [5, 4, 5, 5, 5]
}

for name, prompt in prompts.items():
    average = sum(scores[name]) / len(criteria)
    print("\n", name)
    print("Prompt:", prompt.strip())
    print("Scores:", scores[name])
    print("Average:", round(average, 2), "/ 5")

best = max(scores, key=lambda p: sum(scores[p]))
```

```
print("\nBest prompt:", best)
print("Refined prompt:")
print("""
Explain three applications of AI in healthcare.
For each application, provide a brief explanation
and one realistic example. Use simple language,
a numbered list, and approximately 150 words.
Avoid unsupported medical claims.
""")
```

OUTPUT:

```
 Prompt A
Prompt: Explain AI in healthcare.
Scores: [4, 4, 3, 4, 3]
Average: 3.6 / 5

 Prompt B
Prompt: Explain three applications of AI in healthcare.
Give one example for each application.
Use simple language and a numbered list.
Scores: [5, 4, 5, 5, 5]
Average: 4.8 / 5

Best prompt: Prompt B
Refined prompt:

Explain three applications of AI in healthcare.
For each application, provide a brief explanation
and one realistic example. Use simple language,
a numbered list, and approximately 150 words.
Avoid unsupported medical claims.
```